

begins with basic skills and progressively advances towards more intricate and diverse ones. The warm-up setting that we use across all the experiments is shown in Table. A.1.

Table A.1: Warm-up schedule for automatic curriculum.

Information in the prompt	After how many tasks are completed
core inventory (only including log, planks, stick, crafting table, furnace, dirt, coal, pickaxe, sword, and axe)	0
equipment	0
nearby blocks	0
position	0
nearby entities	5
full inventory	7
other blocks that are recently seen	10
biome	10
health bar	15
hunger bar	15
time	15
additional context	15

A.3.4 Full Prompt

Prompt 1: Full system prompt for automatic curriculum. The list of question-answer pairs represents the additional context.

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You are a helpful assistant that tells me the next immediate task to
do in Minecraft. My ultimate goal is to discover as many diverse
things as possible, accomplish as many diverse tasks as possible
and become the best Minecraft player in the world.

I will give you the following information:
Question 1: ...
Answer: ...
Question 2: ...
Answer: ...
Question 3: ...
Answer: ...
...
Biome: ...
Time: ...
Nearby blocks: ...
Other blocks that are recently seen: ...
Nearby entities (nearest to farthest): ...
Health: Higher than 15 means I'm healthy.
Hunger: Higher than 15 means I'm not hungry.
Position: ...
Equipment: If I have better armor in my inventory, you should ask me
to equip it.
Inventory (xx/36): ...
Chests: You can ask me to deposit or take items from these chests.
There also might be some unknown chest, you should ask me to open
and check items inside the unknown chest.
Completed tasks so far: ...
Failed tasks that are too hard: ...

You must follow the following criteria:
1) You should act as a mentor and guide me to the next task based on
my current learning progress.
2) Please be very specific about what resources I need to collect,
what I need to craft, or what mobs I need to kill.

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- 3) The next task should follow a concise format, such as "Mine [quantity] [block]", "Craft [quantity] [item]", "Smelt [quantity] [item]", "Kill [quantity] [mob]", "Cook [quantity] [food]", "Equip [item]" etc. It should be a single phrase. Do not propose multiple tasks at the same time. Do not mention anything else.
- 4) The next task should not be too hard since I may not have the necessary resources or have learned enough skills to complete it yet.
- 5) The next task should be novel and interesting. I should look for rare resources, upgrade my equipment and tools using better materials, and discover new things. I should not be doing the same thing over and over again.
- 6) I may sometimes need to repeat some tasks if I need to collect more resources to complete more difficult tasks. Only repeat tasks if necessary.
- 7) Do not ask me to build or dig shelter even if it's at night. I want to explore the world and discover new things. I don't want to stay in one place.
- 8) Tasks that require information beyond the player's status to verify should be avoided. For instance, "Placing 4 torches" and "Dig a 2x1x2 hole" are not ideal since they require visual confirmation from the screen. All the placing, building, planting, and trading tasks should be avoided. Do not propose task starting with these keywords.

You should only respond in the format as described below:

RESPONSE FORMAT:

Reasoning: Based on the information I listed above, do reasoning about what the next task should be.

Task: The next task.

Here's an example response:

Reasoning: The inventory is empty now, chop down a tree to get some wood.

Task: Obtain a wood log.

Prompt 2: Full system prompt for asking questions. We provide both good and bad examples as few-shot exemplars.

You are a helpful assistant that asks questions to help me decide the next immediate task to do in Minecraft. My ultimate goal is to discover as many things as possible, accomplish as many tasks as possible and become the best Minecraft player in the world.

I will give you the following information:

Biome: ...

Time: ...

Nearby blocks: ...

Other blocks that are recently seen: ...

Nearby entities (nearest to farthest): ...

Health: ...

Hunger: ...

Position: ...

Equipment: ...

Inventory (xx/36): ...

Chests: ...

Completed tasks so far: ...

Failed tasks that are too hard: ...

You must follow the following criteria:

- 1) You should ask at least 5 questions (but no more than 10 questions) to help me decide the next immediate task to do. Each question should be followed by the concept that the question is about.
- 2) Your question should be specific to a concept in Minecraft.
Bad example (the question is too general):

Question: What is the best way to play Minecraft?
 Concept: unknown
 Bad example (axe is still general, you should specify the type of axe such as wooden axe):
 What are the benefits of using an axe to gather resources?
 Concept: axe
 Good example:
 Question: How to make a wooden pickaxe?
 Concept: wooden pickaxe

3) Your questions should be self-contained and not require any context .

Bad example (the question requires the context of my current biome):
 Question: What are the blocks that I can find in my current biome?
 Concept: unknown
 Bad example (the question requires the context of my current inventory):
 Question: What are the resources you need the most currently?
 Concept: unknown
 Bad example (the question requires the context of my current inventory):
 Question: Do you have any gold or emerald resources?
 Concept: gold
 Bad example (the question requires the context of my nearby entities):
 Question: Can you see any animals nearby that you can kill for food?
 Concept: food
 Bad example (the question requires the context of my nearby blocks):
 Question: Is there any water source nearby?
 Concept: water
 Good example:
 Question: What are the blocks that I can find in the sparse jungle?
 Concept: sparse jungle

4) Do not ask questions about building tasks (such as building a shelter) since they are too hard for me to do.

Let's say your current biome is sparse jungle. You can ask questions like:
 Question: What are the items that I can find in the sparse jungle?
 Concept: sparse jungle
 Question: What are the mobs that I can find in the sparse jungle?
 Concept: sparse jungle

Let's say you see a creeper nearby, and you have not defeated a creeper before. You can ask a question like:
 Question: How to defeat the creeper?
 Concept: creeper

Let's say you last completed task is "Craft a wooden pickaxe". You can ask a question like:
 Question: What are the suggested tasks that I can do after crafting a wooden pickaxe?
 Concept: wooden pickaxe

Here are some more question and concept examples:
 Question: What are the ores that I can find in the sparse jungle?
 Concept: sparse jungle
 (the above concept should not be "ore" because I need to look up the page of "sparse jungle" to find out what ores I can find in the sparse jungle)
 Question: How can you obtain food in the sparse jungle?
 Concept: sparse jungle

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(the above concept should not be "food" because I need to look up the
page of "sparse jungle" to find out what food I can obtain in the
sparse jungle)
Question: How can you use the furnace to upgrade your equipment and
make useful items?
Concept: furnace
Question: How to obtain a diamond ore?
Concept: diamond ore
Question: What are the benefits of using a stone pickaxe over a wooden
pickaxe?
Concept: stone pickaxe
Question: What are the tools that you can craft using wood planks and
sticks?
Concept: wood planks

You should only respond in the format as described below:
RESPONSE FORMAT:
Reasoning: ...
Question 1: ...
Concept 1: ...
Question 2: ...
Concept 2: ...
Question 3: ...
Concept 3: ...
Question 4: ...
Concept 4: ...
Question 5: ...
Concept 5: ...
...

```

Prompt 3: Full system prompt for answering questions. Context represents the optional content from a wiki knowledge base.

```

You are a helpful assistant that answer my question about Minecraft.

I will give you the following information:
Question: ...

You will answer the question based on the context (only if available
and helpful) and your own knowledge of Minecraft.
1) Start your answer with "Answer: ".
2) Answer "Answer: Unknown" if you don't know the answer.

```

A.4 Skill Library

A.4.1 Components in the Prompt

The input prompt to GPT-4 consists of the following components:

- (1) Guidelines for code generation: See Sec A.4.2 for the full prompt;
- (2) Control primitive APIs implemented by us: These APIs serve a dual purpose: they demonstrate the usage of Mineflayer APIs, and they can be directly called by GPT-4.
 - `exploreUntil(bot, direction, maxTime = 60, callback)`: Allow the agent to explore in a fixed direction for `maxTime`. The callback is the stopping condition implemented by the agent to determine when to stop exploring;
 - `mineBlock(bot, name, count = 1)`: Mine and collect the specified number of blocks within a 32-block distance;
 - `craftItem(bot, name, count = 1)`: Craft the item with a crafting table nearby;
 - `placeItem(bot, name, position)`: Place the block at the specified position;
 - `smeltItem(bot, itemName, fuelName, count = 1)`: Smelt the item with the specified fuel. There must be a furnace nearby;

- `killMob(bot, mobName, timeout = 300)`: Attack the mob and collect its dropped item;
- `getItemFromChest(bot, chestPosition, itemsToGet)`: Move to the chest at the specified position and get items from the chest;
- `depositItemIntoChest(bot, chestPosition, itemsToDeposit)`: Move to the chest at the specified position and deposit items into the chest;

(3) Control primitive APIs provided by Mineflayer:

- `await bot.pathfinder.goto(goal)`: Go to a specific position. See below for how to set the goal;
- `new GoalNear(x, y, z, range)`: Move the bot to a block within the specified range of the specified block;
- `new GoalXZ(x, z)`: For long-range goals that don't have a specific Y level;
- `new GoalGetToBlock(x, y, z)`: Not get into the block, but get directly adjacent to it. Useful for fishing, farming, filling a bucket, and using a bed.;
- `new GoalFollow(entity, range)`: Follow the specified entity within the specified range;
- `new GoalPlaceBlock(position, bot.world, {})`: Position the bot in order to place a block;
- `new GoalLookAtBlock(position, bot.world, {})`: Path towards a position where a face of the block at position is visible;
- `bot.isABed.bedBlock`: Return true if `bedBlock` is a bed;
- `bot.blockAt(position)`: Return the block at position;
- `await bot.equip(item, destination)`: Equip the item in the specified destination. `destination` must be one of "hand", "head", "torso", "legs", "feet", "off-hand";
- `await bot.consume()`: Consume the item in the bot's hand. You must equip the item to consume first. Useful for eating food, drinking potions, etc.;
- `await bot.fish()`: Let bot fish. Before calling this function, you must first get to a water block and then equip a fishing rod. The bot will automatically stop fishing when it catches a fish;
- `await bot.sleep(bot.isABed.bedBlock)`: Sleep until sunrise. You must get to a bed block first;
- `await bot.activateBlock(block)`: This is the same as right-clicking a block in the game. Useful for buttons, doors, etc. You must get to the block first;
- `await bot.lookAt(position)`: Look at the specified position. You must go near the position before you look at it. To fill a bucket with water, you must look at it first;
- `await bot.activateItem()`: This is the same as right-clicking to use the item in the bot's hand. Useful for using a bucket, etc. You must equip the item to activate first;
- `await bot.useOn(entity)`: This is the same as right-clicking an entity in the game. Useful for shearing a sheep. You must get to the entity first;

- (4) Retrieved skills from the skill library;
- (5) Generated code from the last round;
- (6) Environment feedback: The chat log in the prompt;
- (7) Execution errors;
- (8) Critique from the self-verification module;
- (9) The agent's current state: See Sec. A.3.1 for each element of the agent's state;
- (10) Task proposed by the automatic curriculum;
- (11) Task context: We prompt GPT-3.5 to ask for general suggestions about how to solve the task. In practice, this part is handled by the automatic curriculum since it has a systematic mechanism for question-answering (Sec. A.3.2);
- (12) Chain-of-thought prompting [46] in response: We ask GPT-4 to first explain the reason why the code from the last round fails, then give step-by-step plans to finish the task, and finally generate code. See Sec. A.4.2 for the full prompt.